

TACTICAL SHELTER THERMAL ASSESSMENT DOSSIER

Theatre: Ladakh High Altitude (34.15N, 77.58E) | Date: 2026-09-02 | Alt: ~4,500m ASL

LOGISTICS PAYLOAD: 19,255 kg (AIRLIFT IMPOSSIBLE: Requires Road Transport (Convoy)) | Thermal Rating: MODERATE BLOOM (+1.90 C)

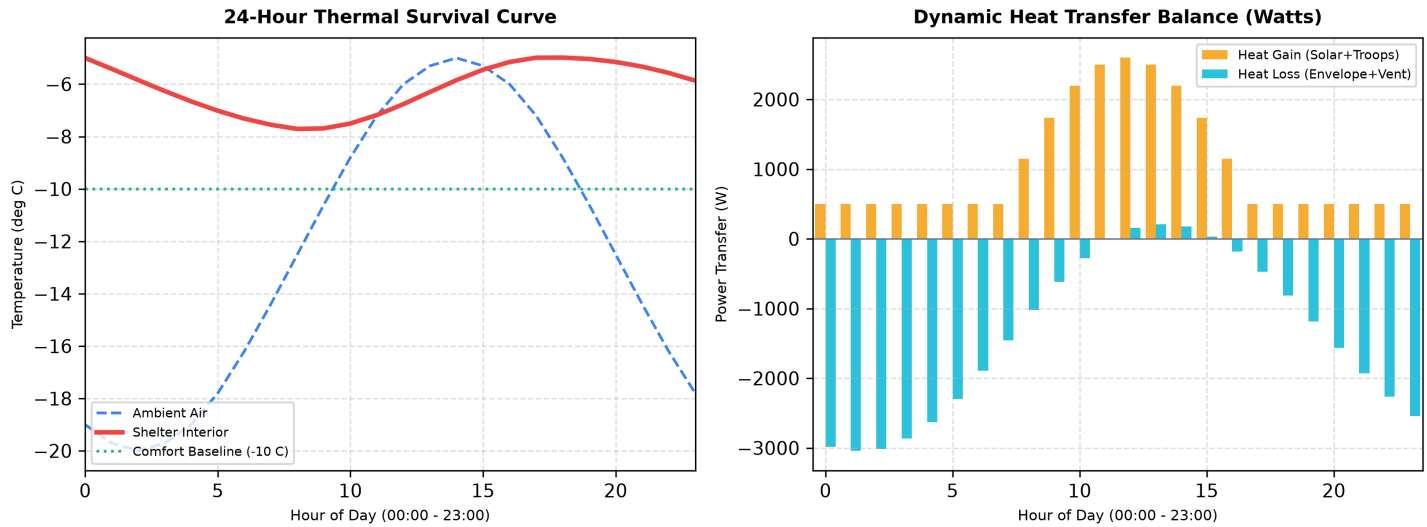
THERMAL SURVIVAL: Min -7.7 C | Max -5.0 C | Avg -6.1 C

TERRAIN SHADOW MAPPING: Disabled

1. Structural Specifications & Biological Heat Matrix

STRUCTURAL PARAMETER	CONFIGURATION	TACTICAL PARAMETER	SPECIFICATION
Dimensions (L x W x H)	6.0m x 4.0m x 2.5m	Active Occupants (Troops)	5 Personnel (500 W Load)
Total Envelope Volume	60.0 m3	Ventilation Rate (ACH)	0.5 Air Changes/Hour
Wall Envelope Material	Brick (R=0.3)	Roof Insulation System	Polyurethane Panel (PU (R=5.0))
Glazing Area & Type	3.0 m2 (Glass (Single Pane))	Est. Procurement Cost	INR 125,500

2. High-Altitude Thermal Dynamics & Heat Flux Curves



3. Tactical Survivability & Deployment Assessment

Under simulated -20.0 C Ladakh operational conditions, the structure maintained an average interior temperature of -6.1 C with 24/24 hours above the -10.0 C survival baseline. Total thermal envelope mass stands at 19,180 kg with a combined deployment weight of 19,255 kg, validating deployment feasibility under: AIRLIFT IMPOSSIBLE: Requires Road Transport (Convoy). Envelope external thermal differential is +1.90 C relative to ambient, fulfilling thermal stealth parameters classified as 'MODERATE BLOOM'.

Engineering Officer Sign-Off: _____

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